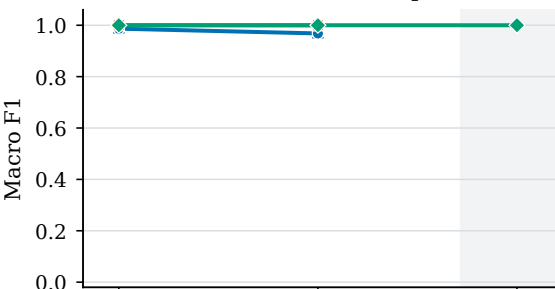


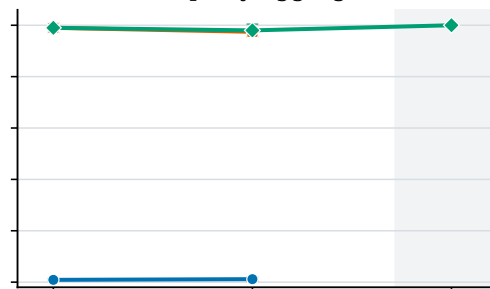
Qwen 3.5

LLM CodeAct RLM

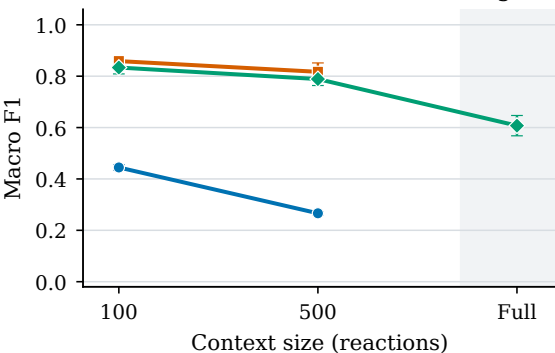
(a) Tier 1: Structural lookup ($n=10$)



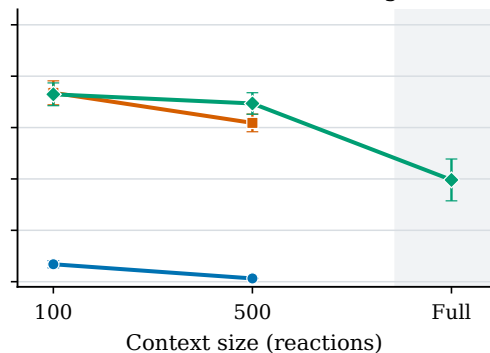
(b) Tier 2: Property aggregation ($n=20$)



(c) Tier 3: Reaction-level reasoning ($n=35$)



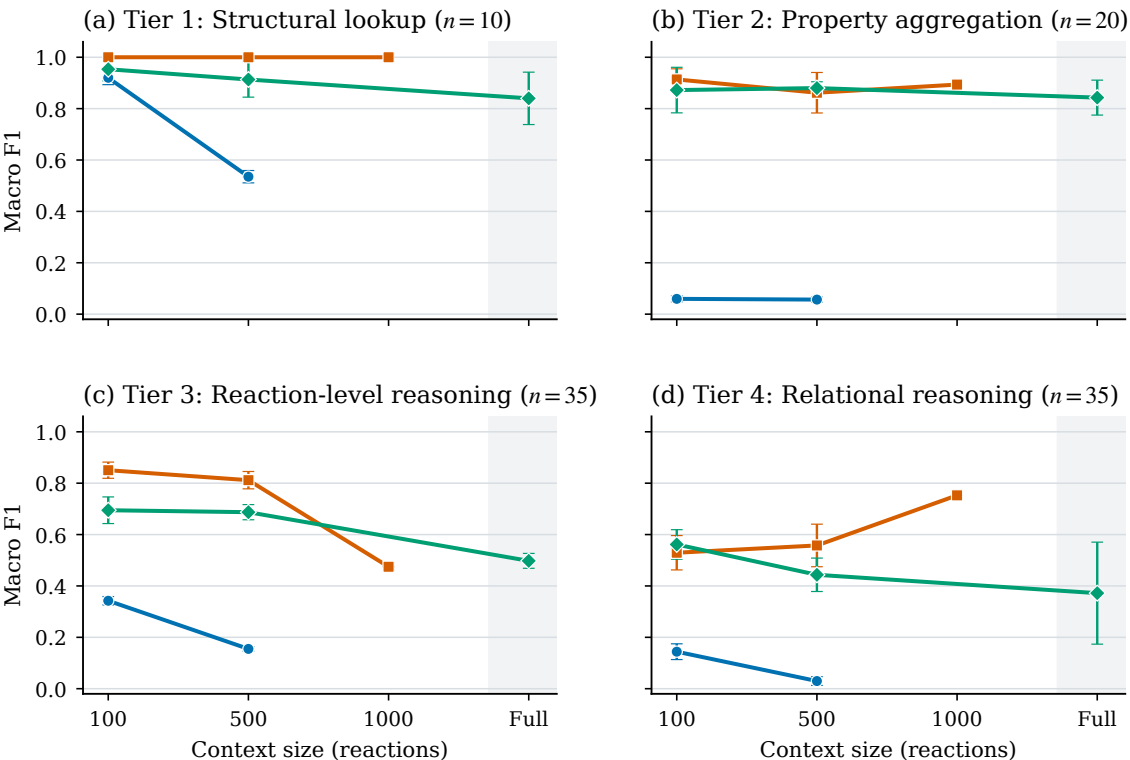
(d) Tier 4: Relational reasoning ($n=35$)



Curves condition on successful runs.

DeepSeek V4 Flash

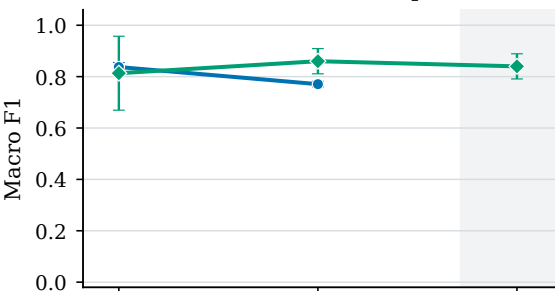
—●— LLM —■— *CodeAct —◆— *RLM



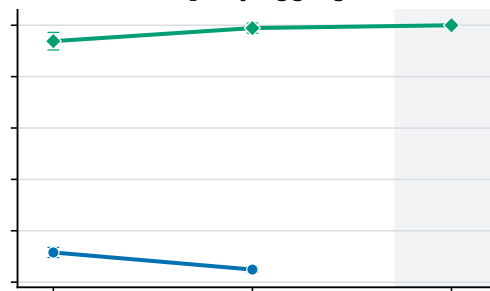
GLM 5.2

—●— LLM —■— *CodeAct —◆— *RLM

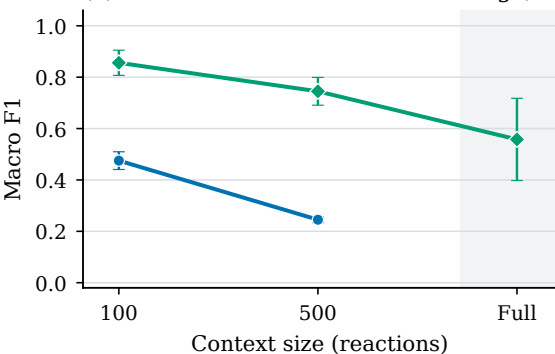
(a) Tier 1: Structural lookup ($n=10$)



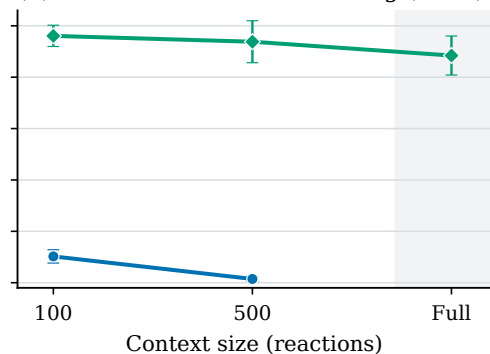
(b) Tier 2: Property aggregation ($n=20$)



(c) Tier 3: Reaction-level reasoning ($n=35$)



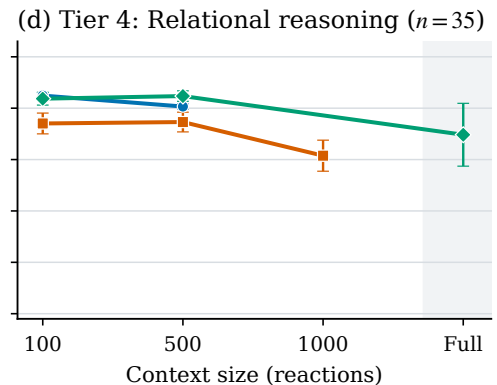
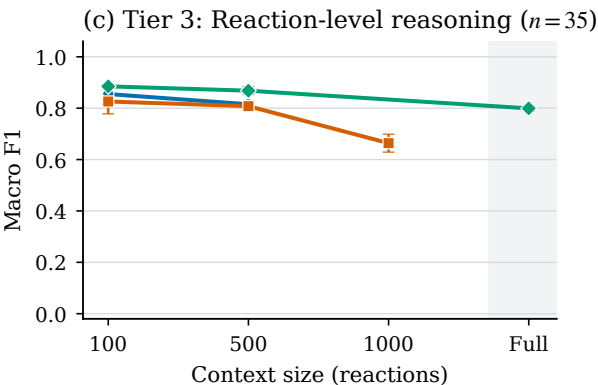
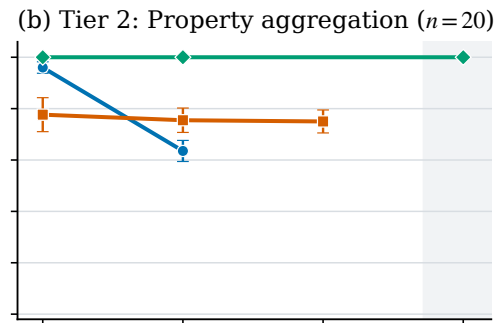
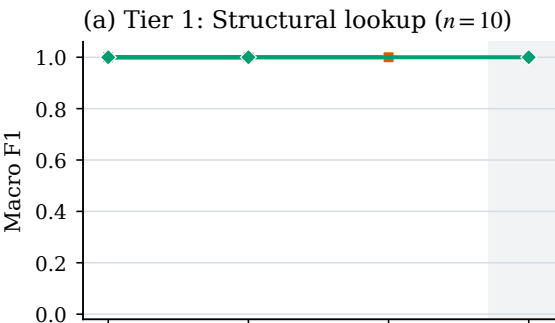
(d) Tier 4: Relational reasoning ($n=35$)



Curves condition on successful runs. * Arm has running, stale, pending, or unrun jobs in the gold snapshot.

Gemini 3.7 Flash

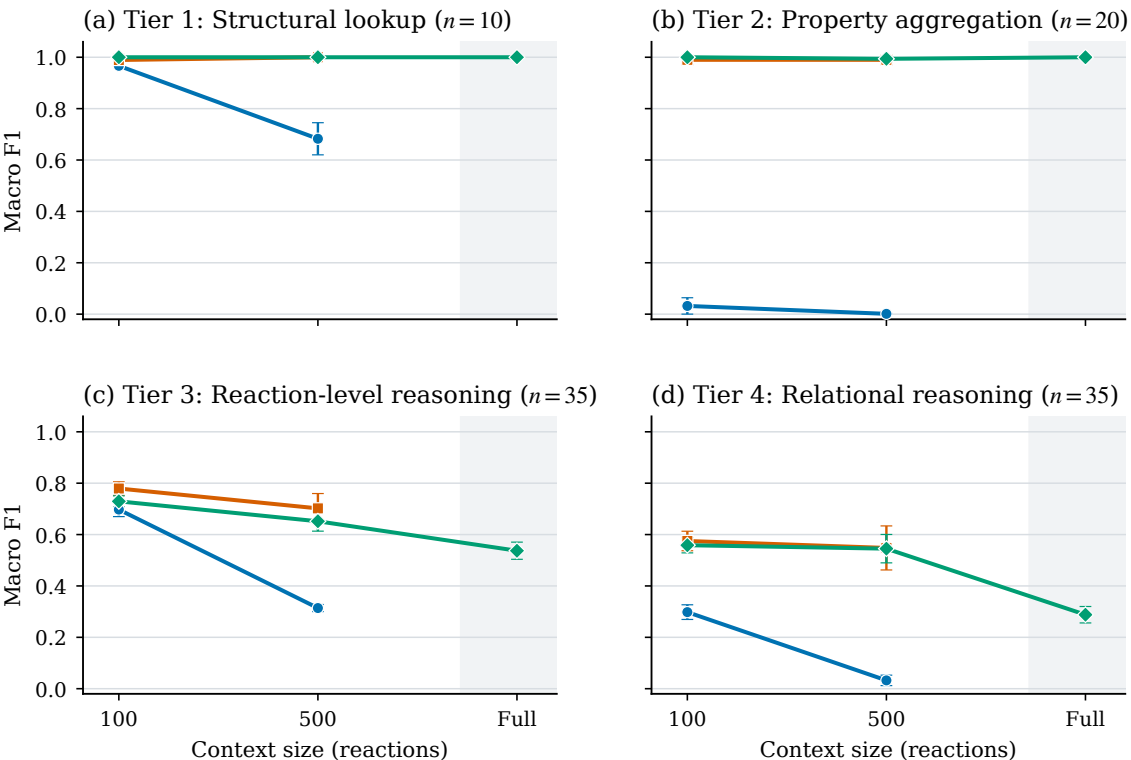
LLM CodeAct RLM



Curves condition on successful runs.

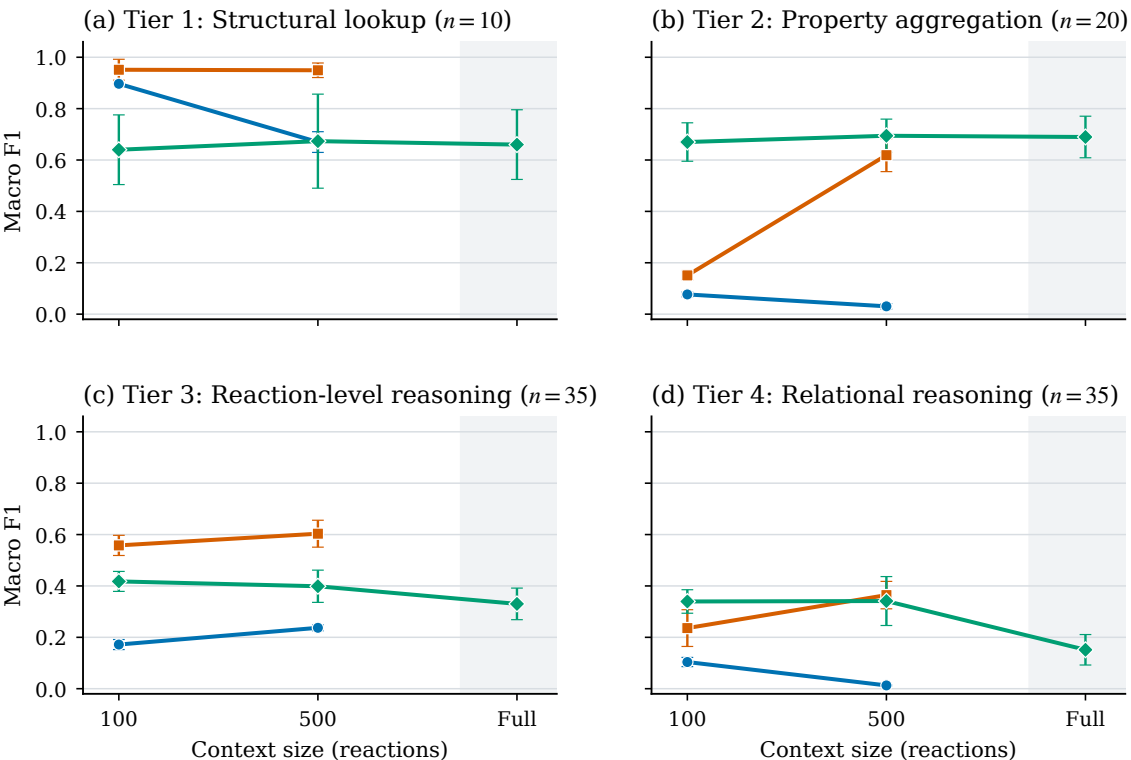
GPT-5 mini

LLM CodeAct RLM



Claude Haiku 4.5

LLM CodeAct RLM



Curves condition on successful runs.